

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

Complete

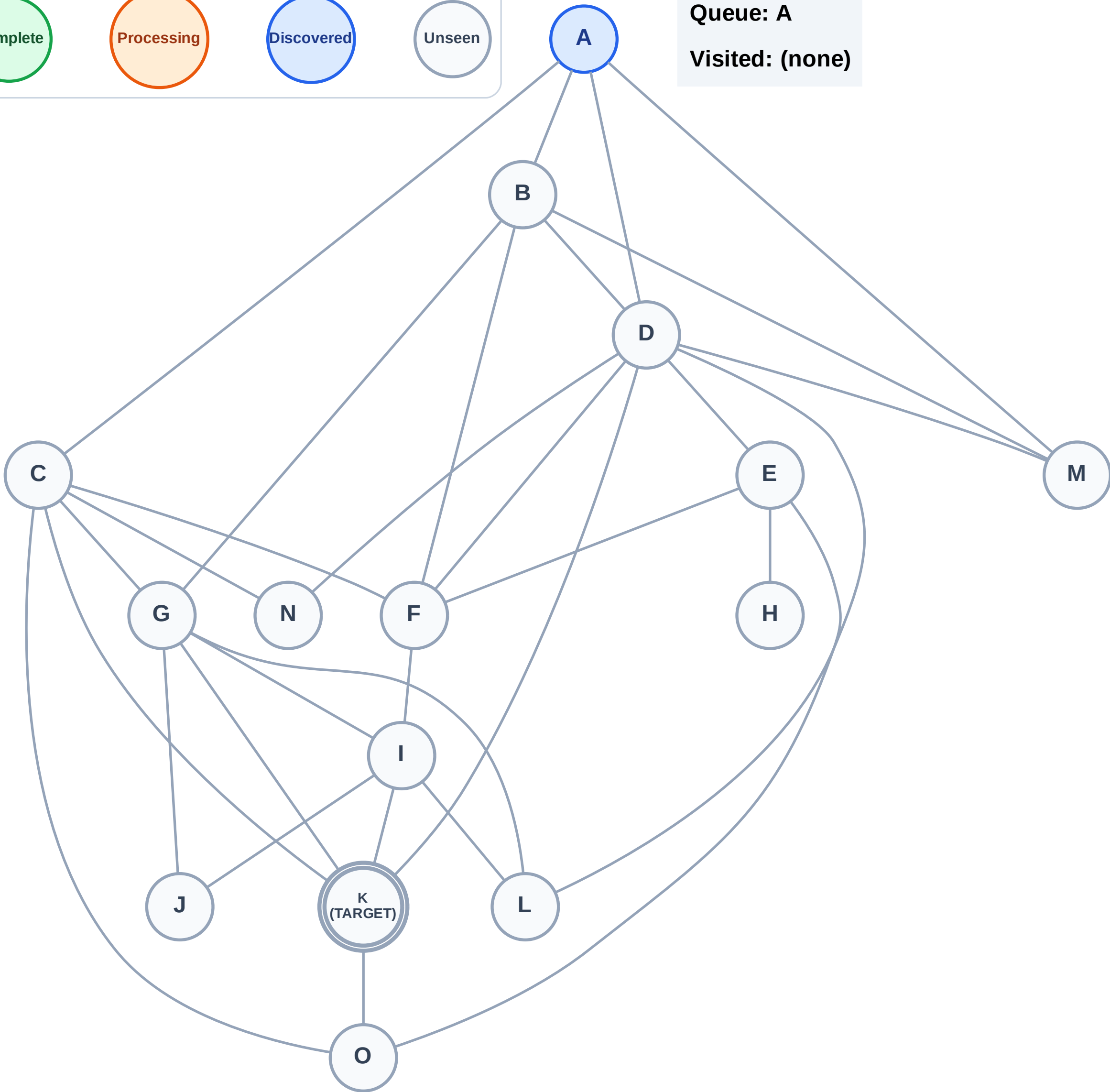
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

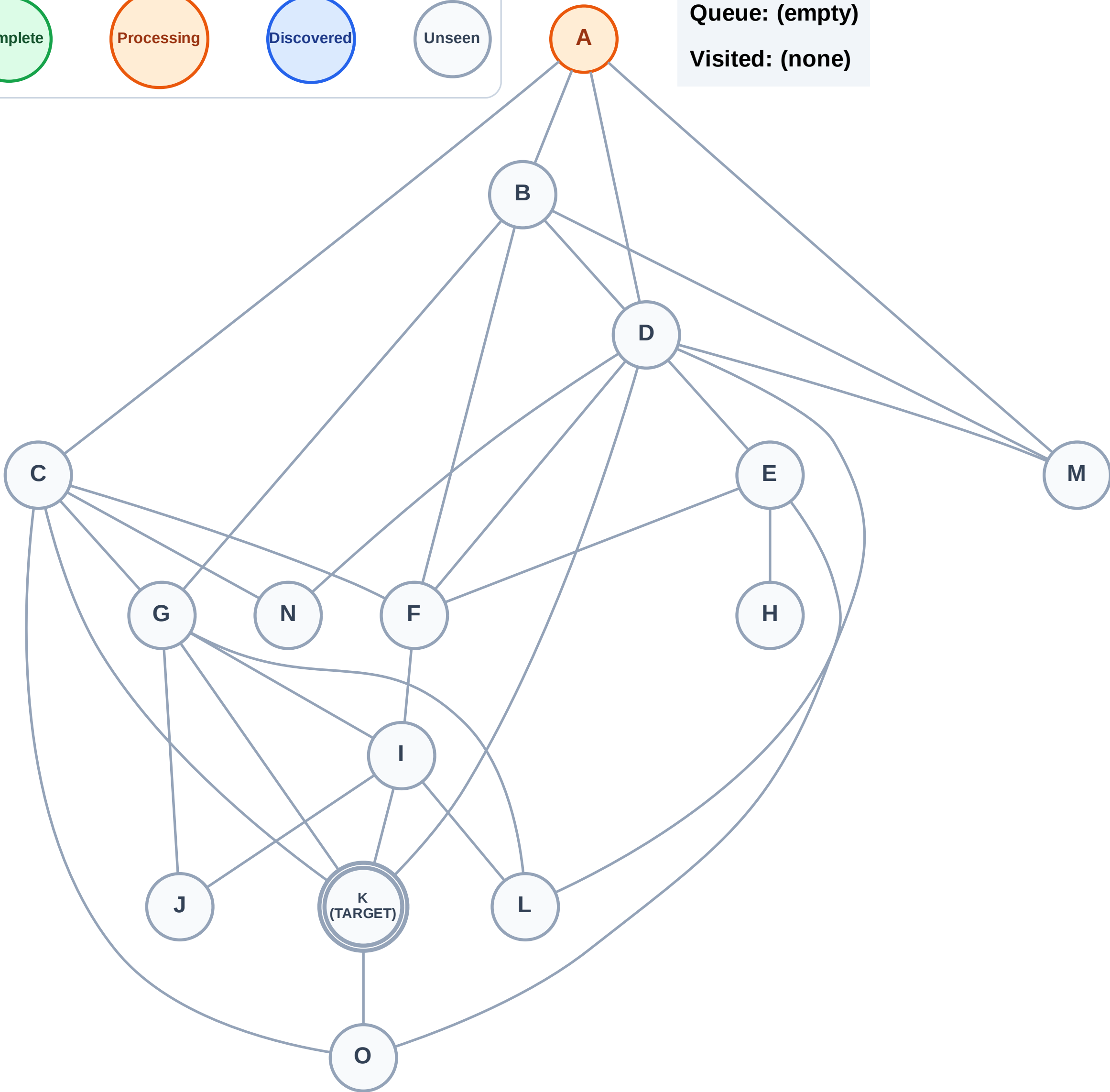
Complete

Processing

Discovered

Unseen

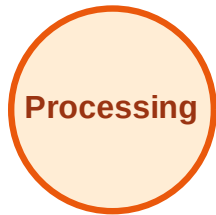
Queue: (empty)
Visited: (none)



BFS Traversal

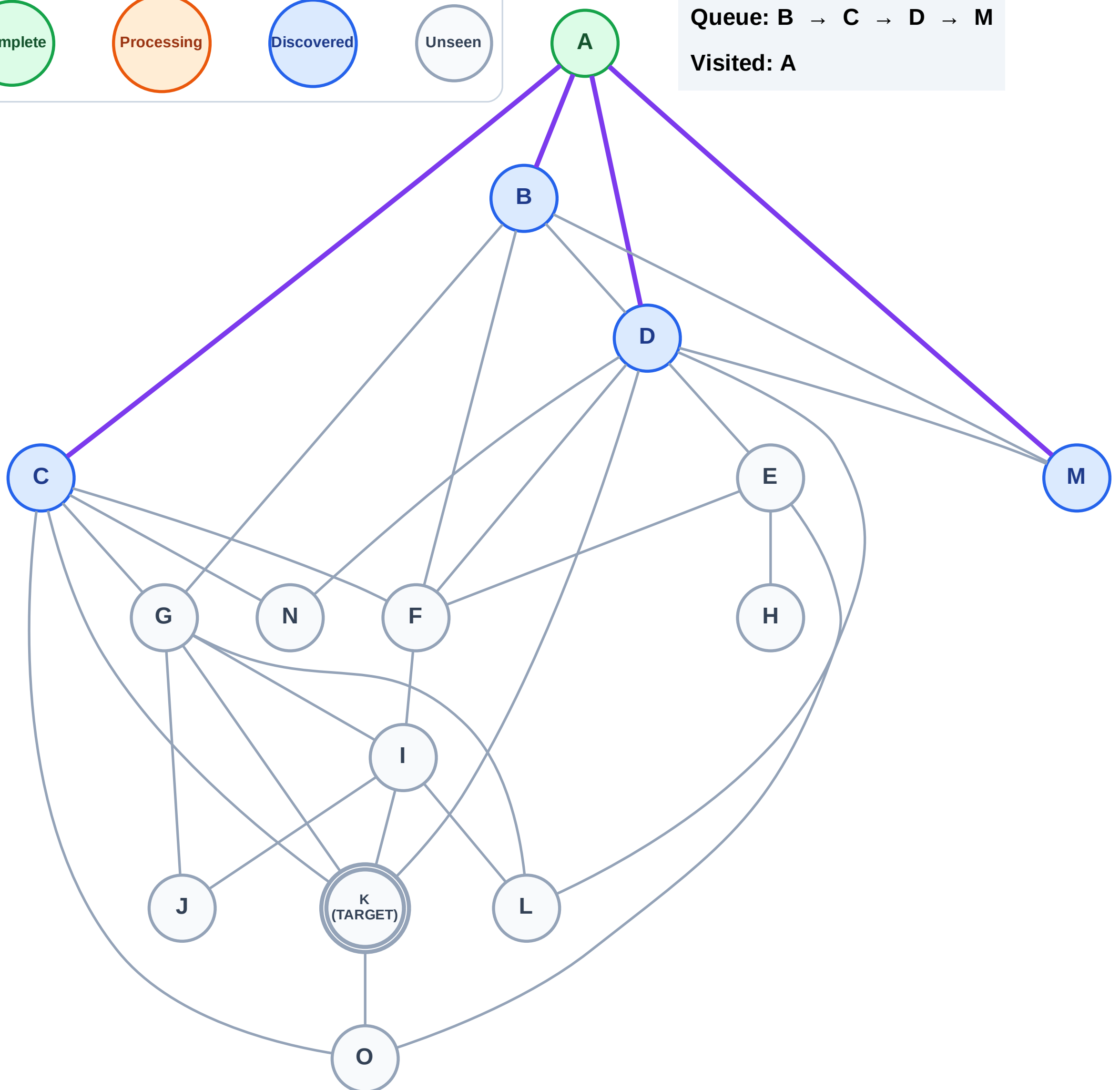
Step 3: Discover B, C, D, M from A

Legend



Queue: B → C → D → M

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

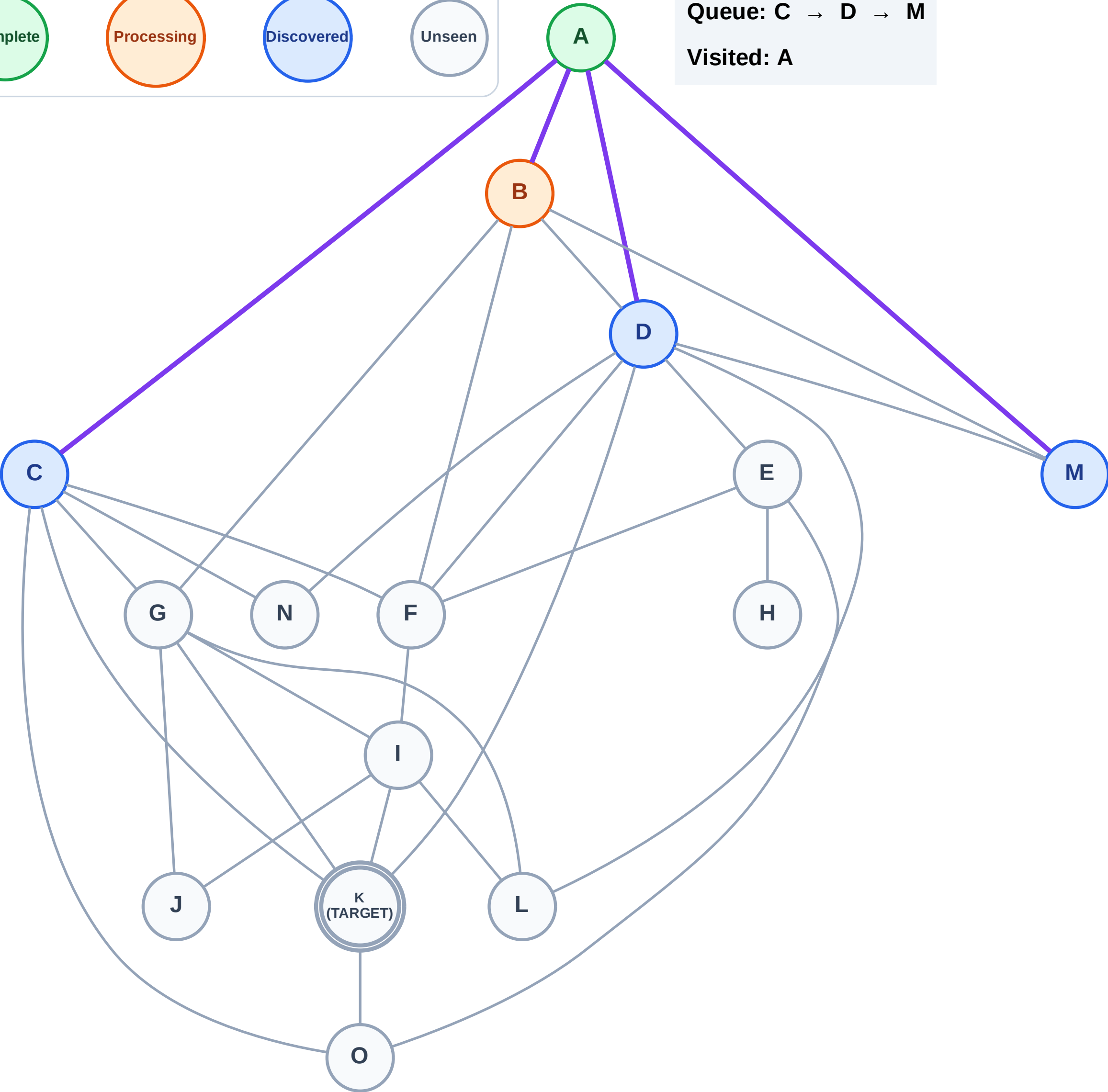
Processing

Discovered

Unseen

Queue: C → D → M

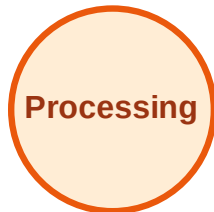
Visited: A



BFS Traversal

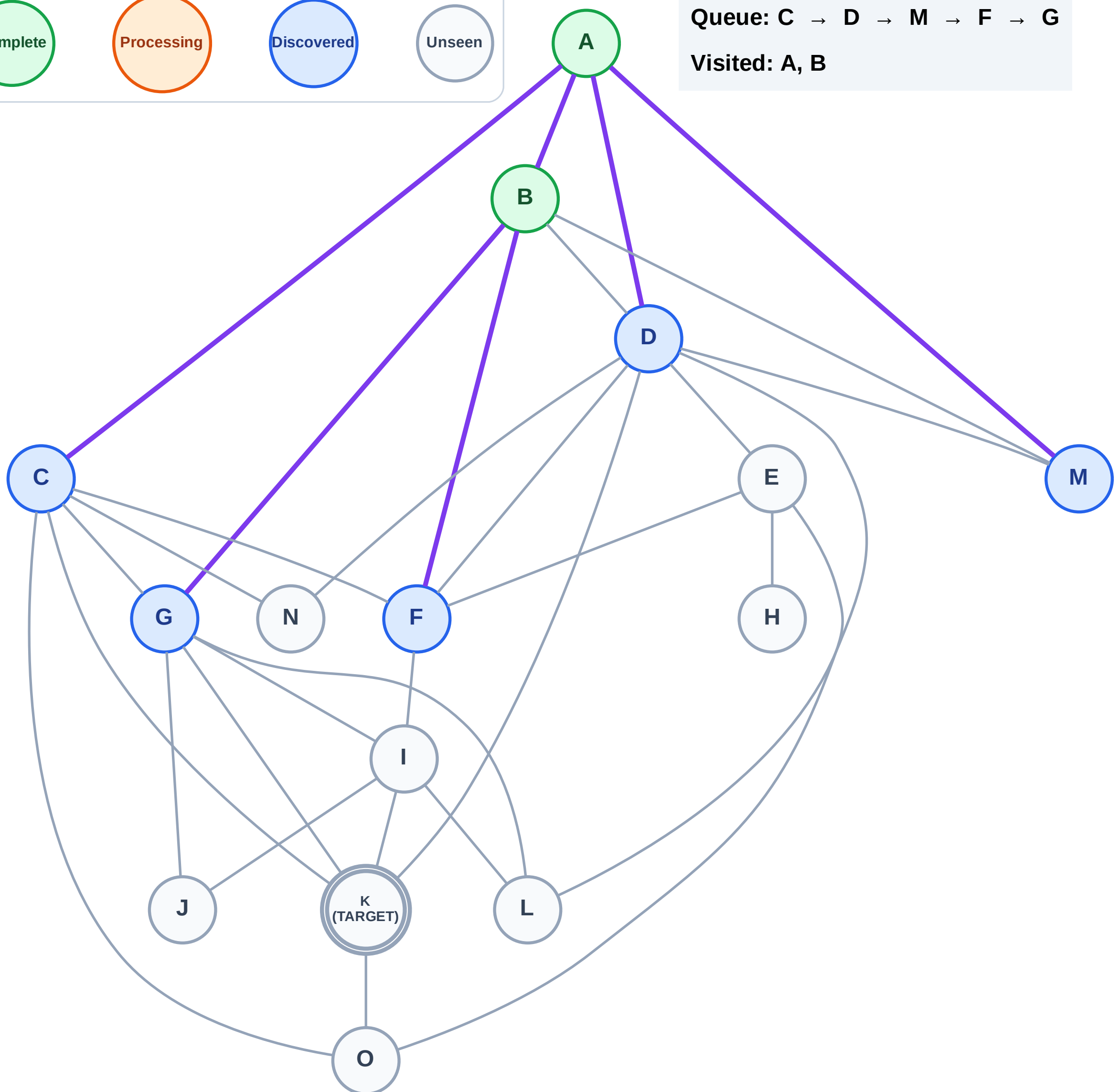
Step 5: Discover F, G from B

Legend



Queue: C → D → M → F → G

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

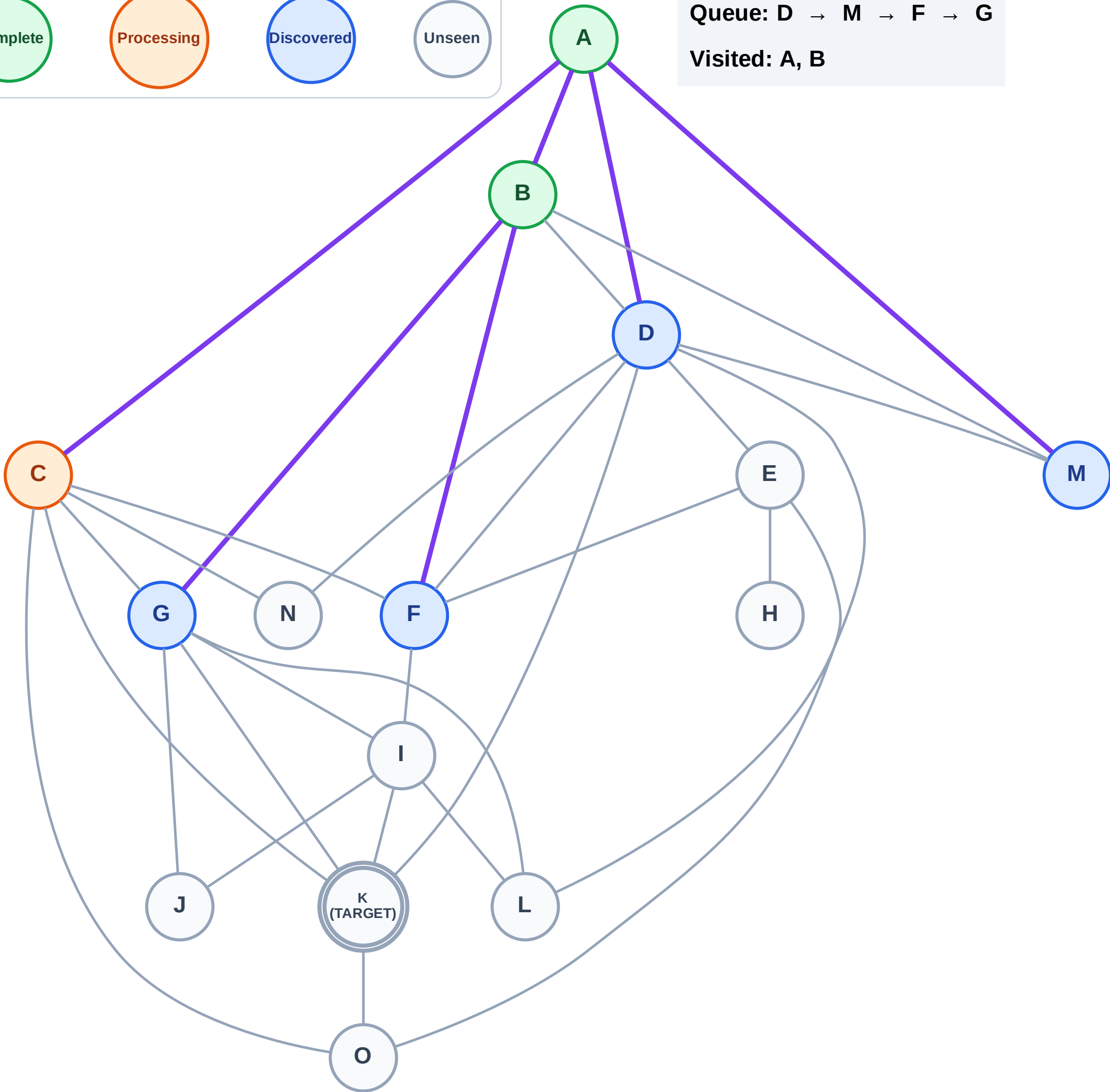
Processing

Discovered

Unseen

Queue: D → M → F → G

Visited: A, B



BFS Traversal

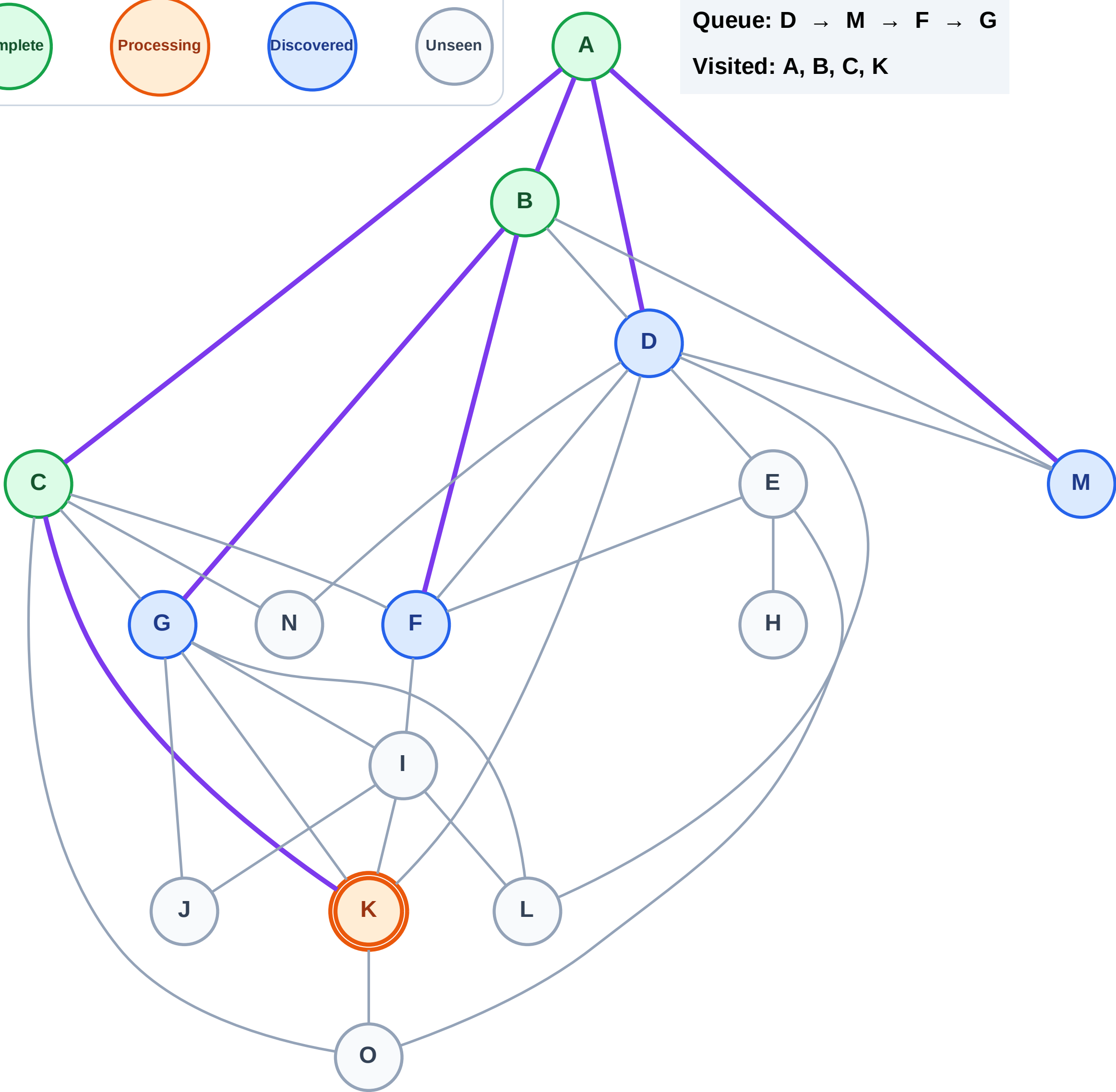
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: D → M → F → G

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

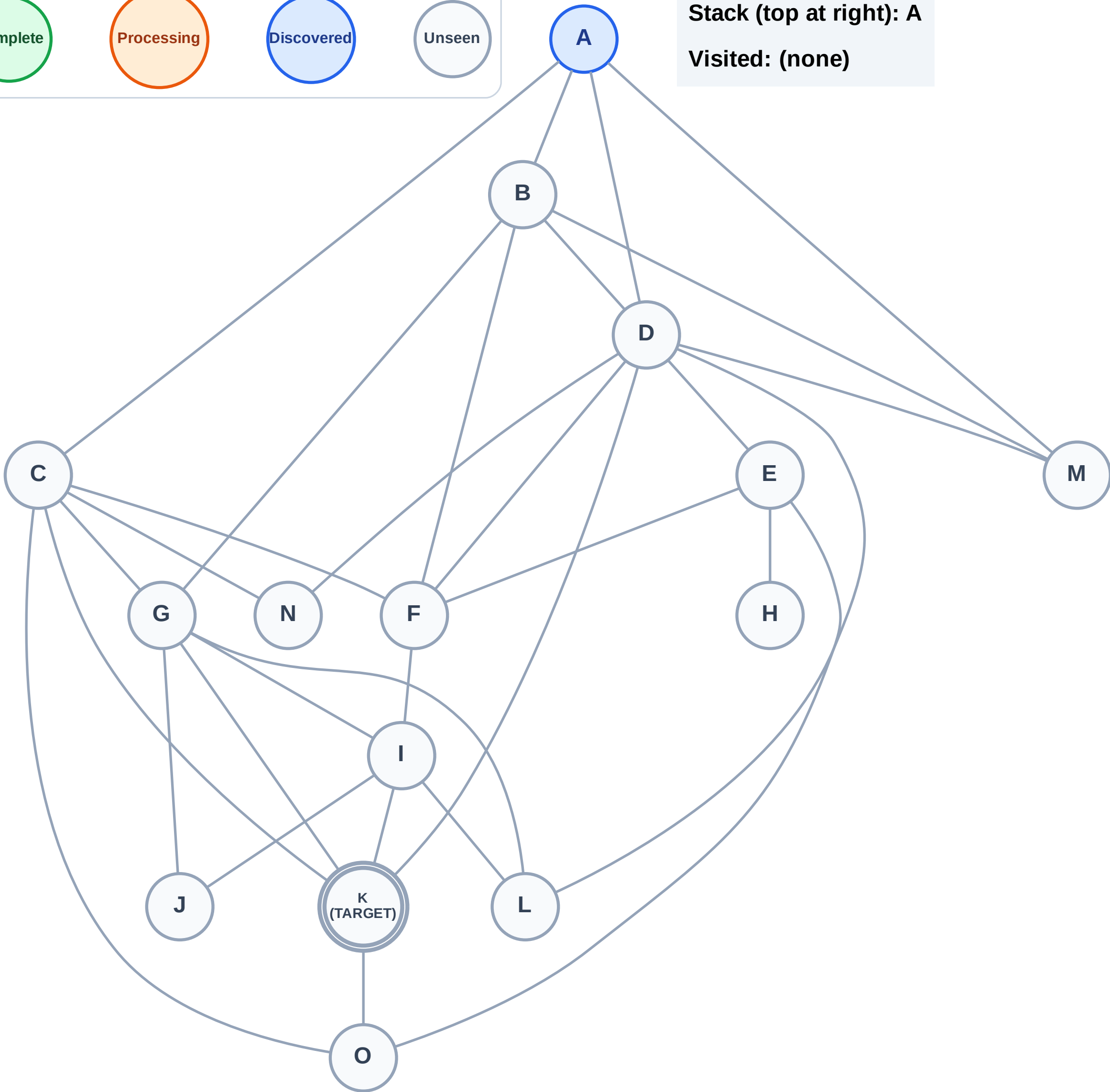
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

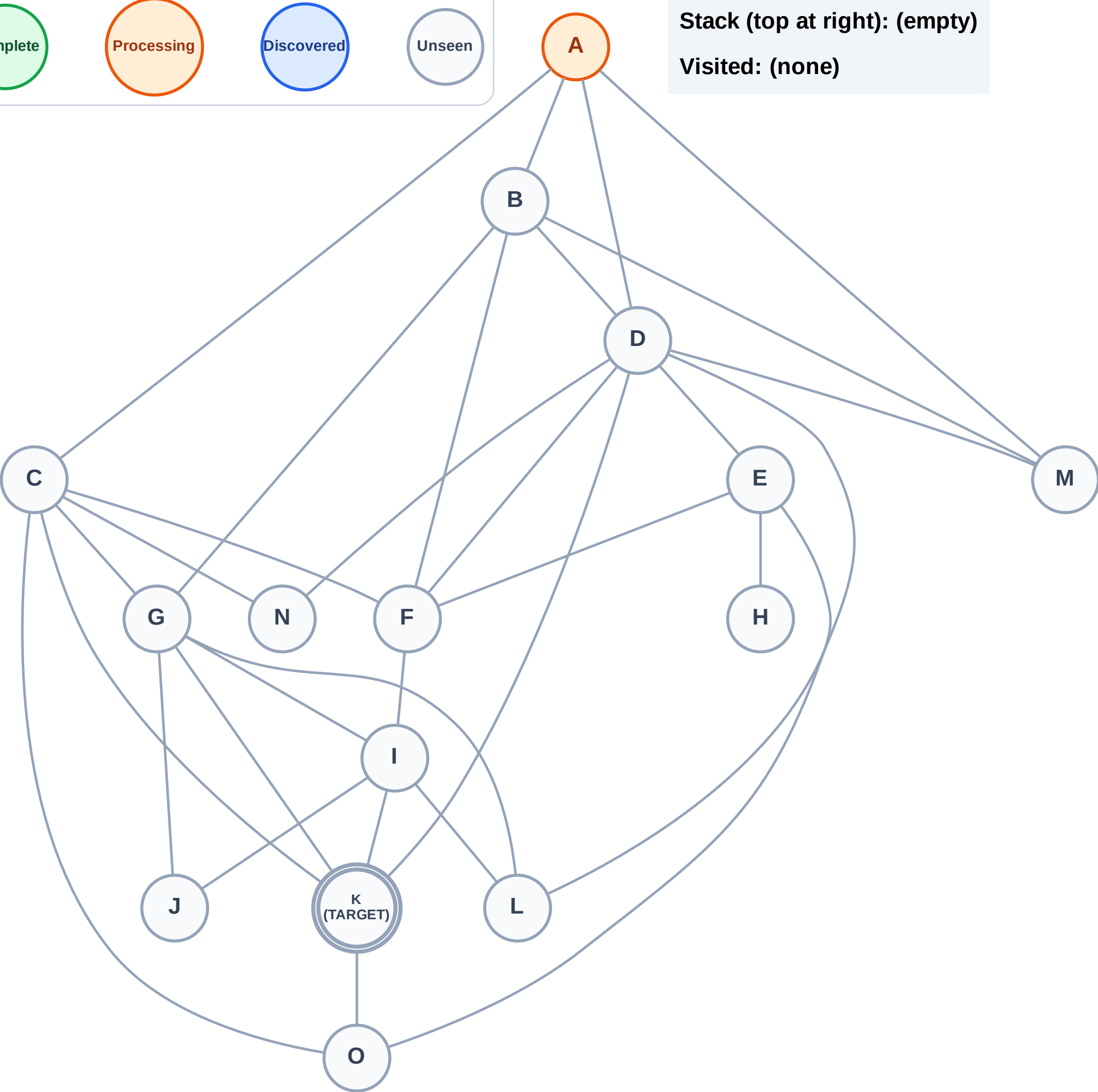
Complete

Processing

Discovered

Unseen

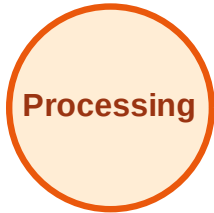
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

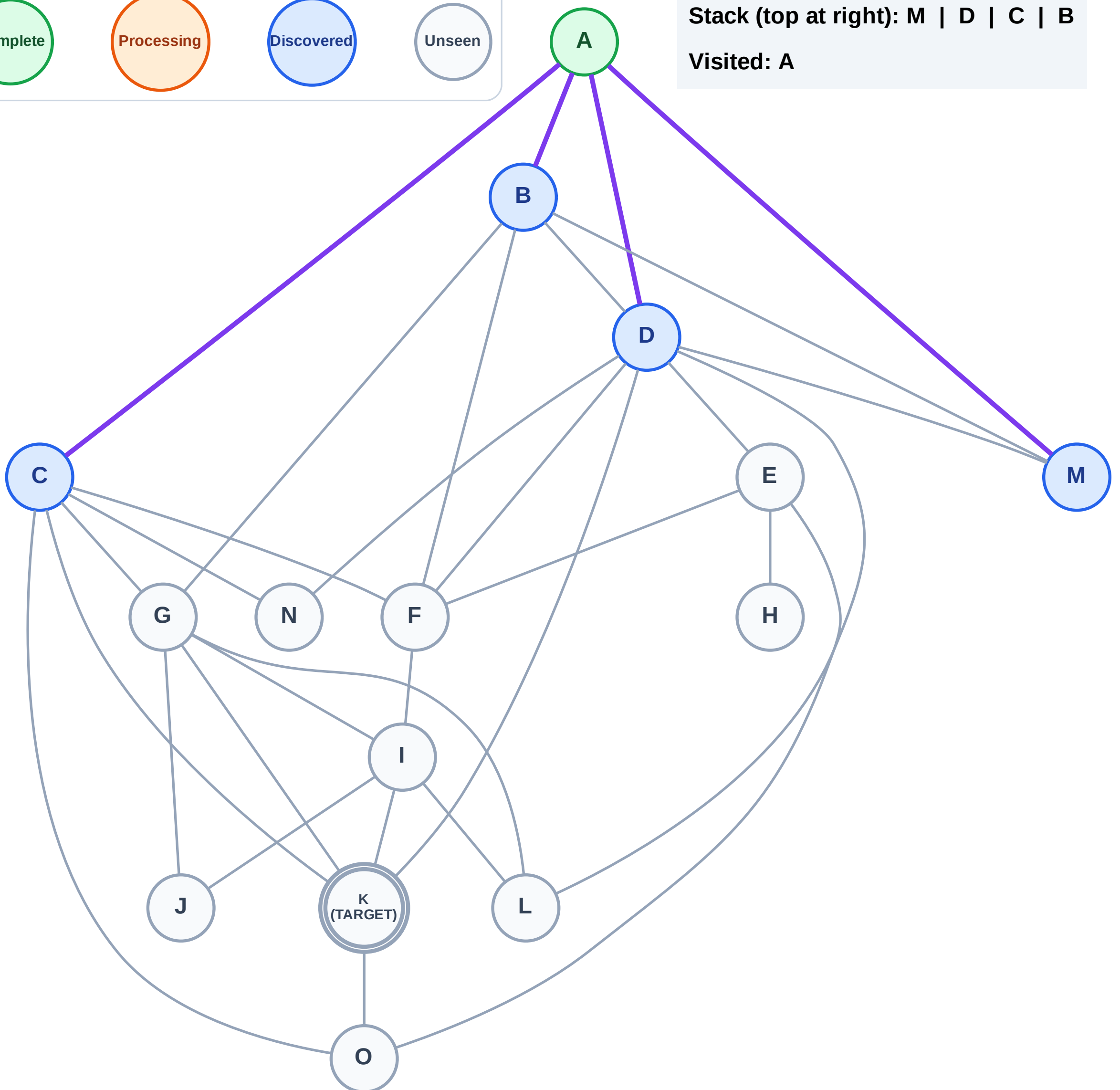
Step 3: Discover M, D, C, B from A

Legend



Stack (top at right): M | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

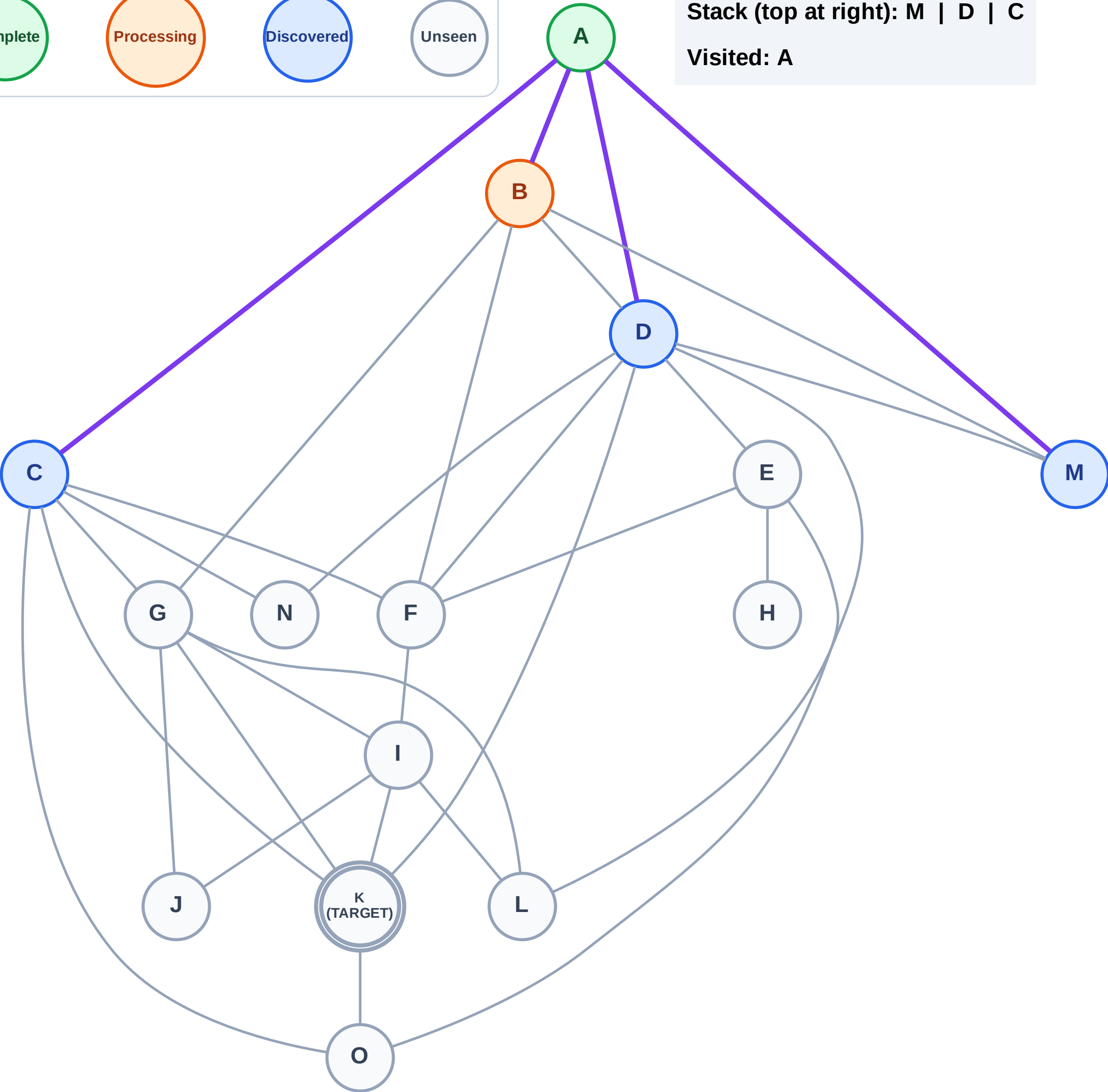
Complete

Processing

Discovered

Unseen

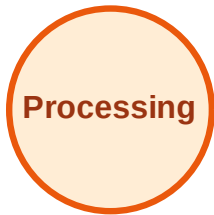
Stack (top at right): M | D | C
Visited: A



DFS Traversal

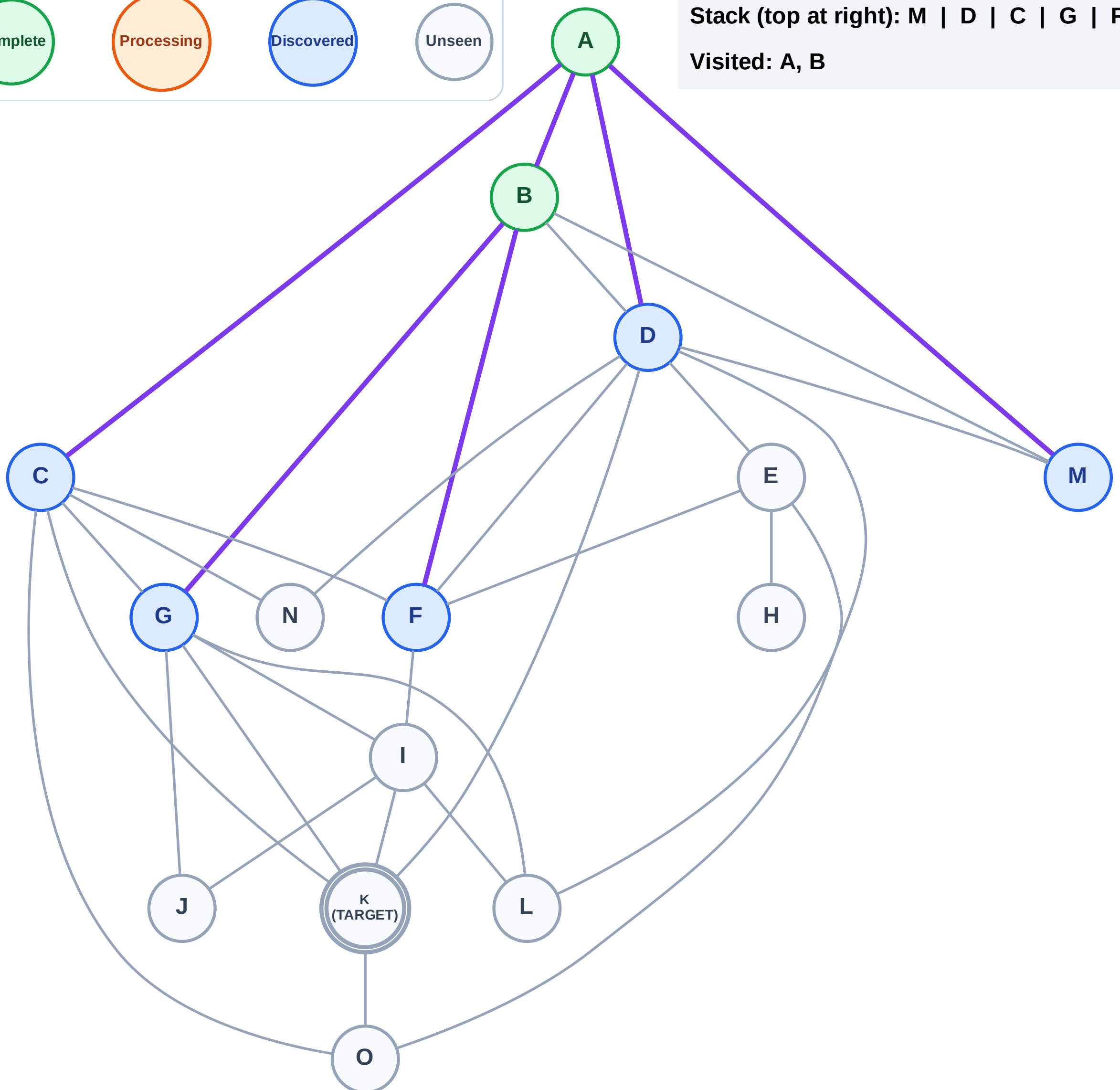
Step 5: Discover G, F from B

Legend



Stack (top at right): M | D | C | G | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

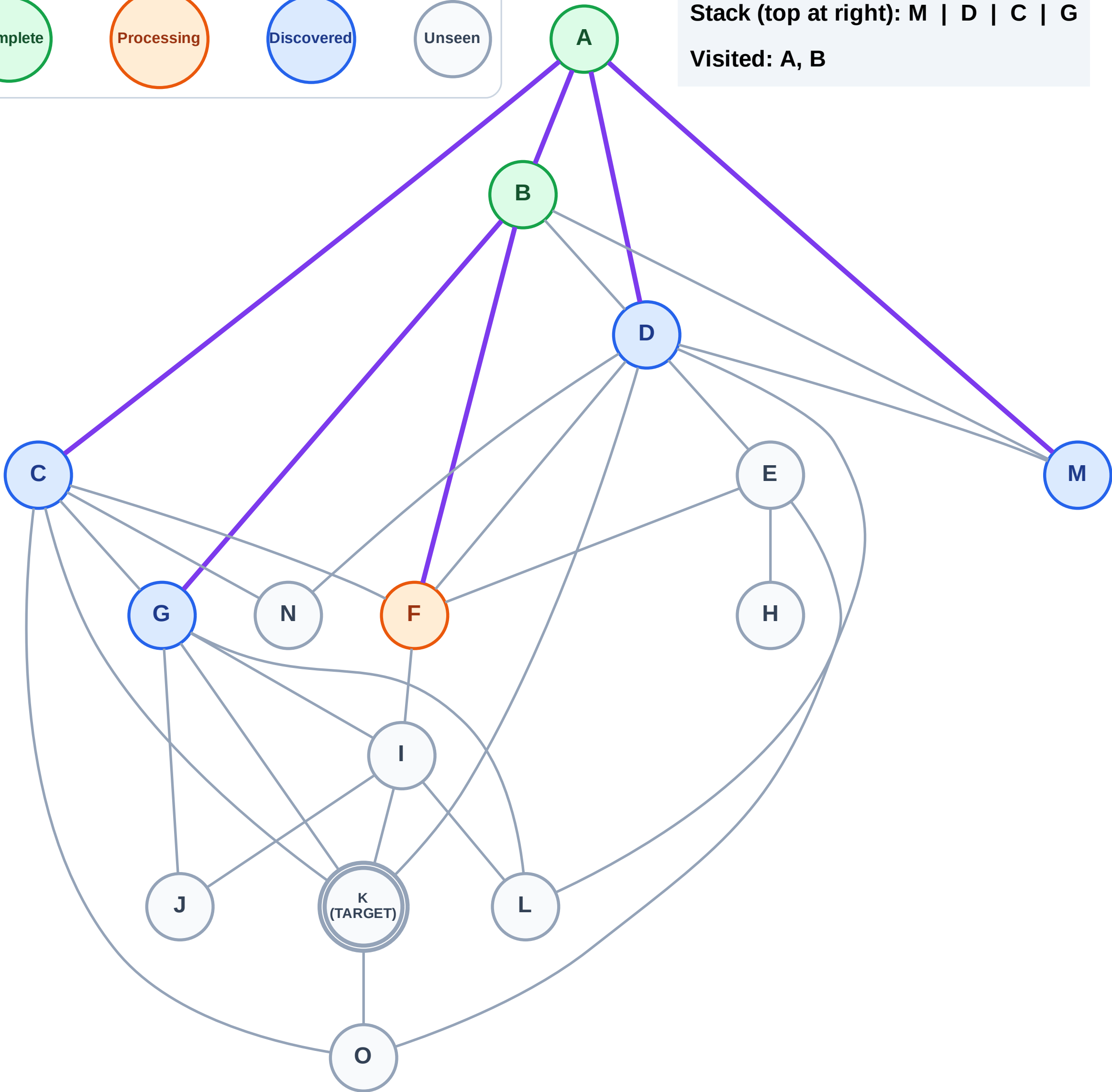
Processing

Discovered

Unseen

Stack (top at right): M | D | C | G

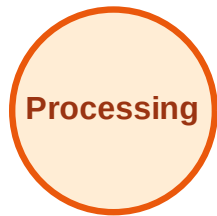
Visited: A, B



DFS Traversal

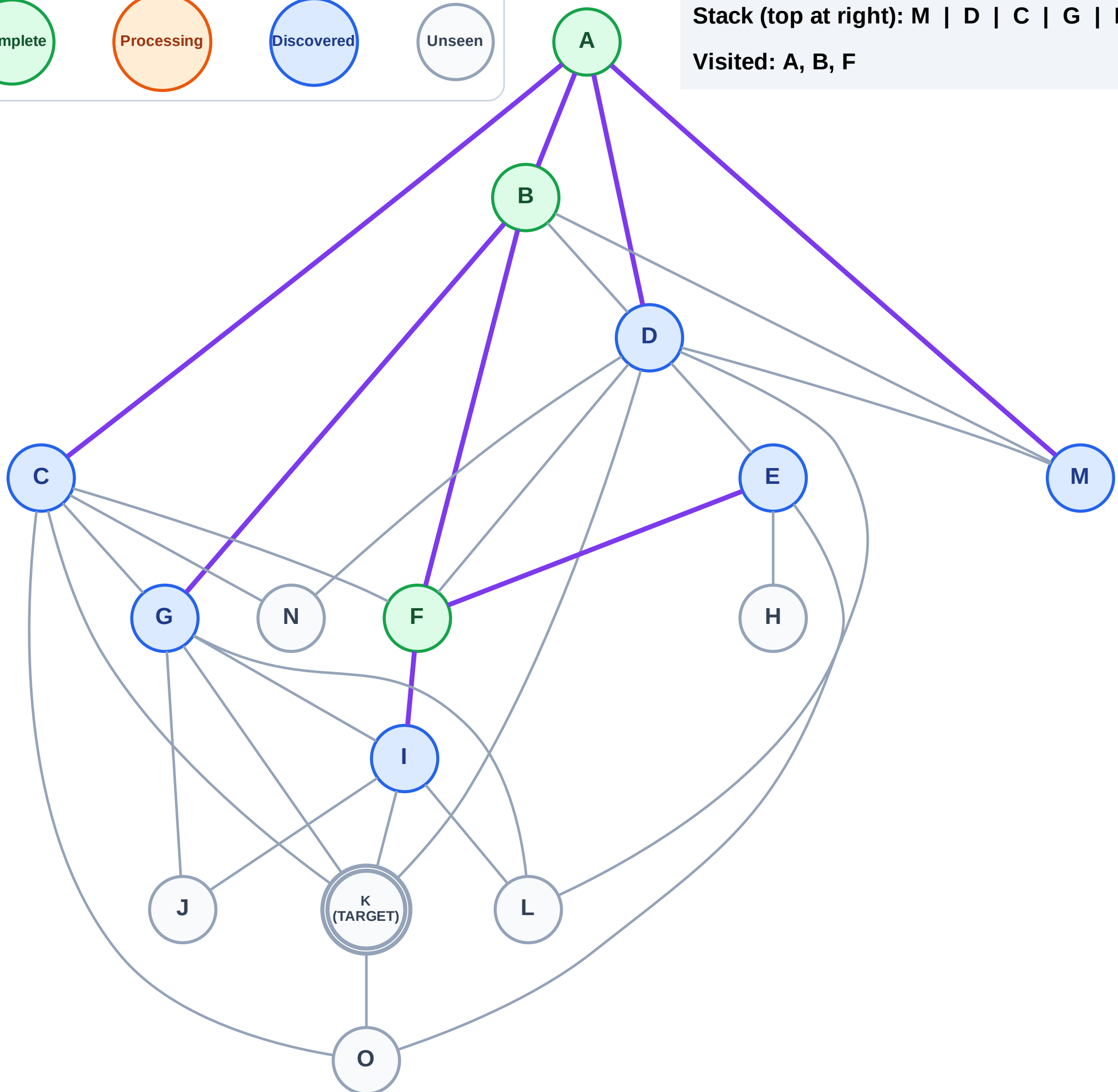
Step 7: Discover I, E from F

Legend



Stack (top at right): M | D | C | G | I | E

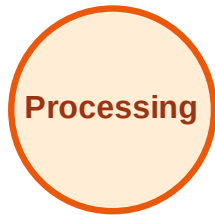
Visited: A, B, F



DFS Traversal

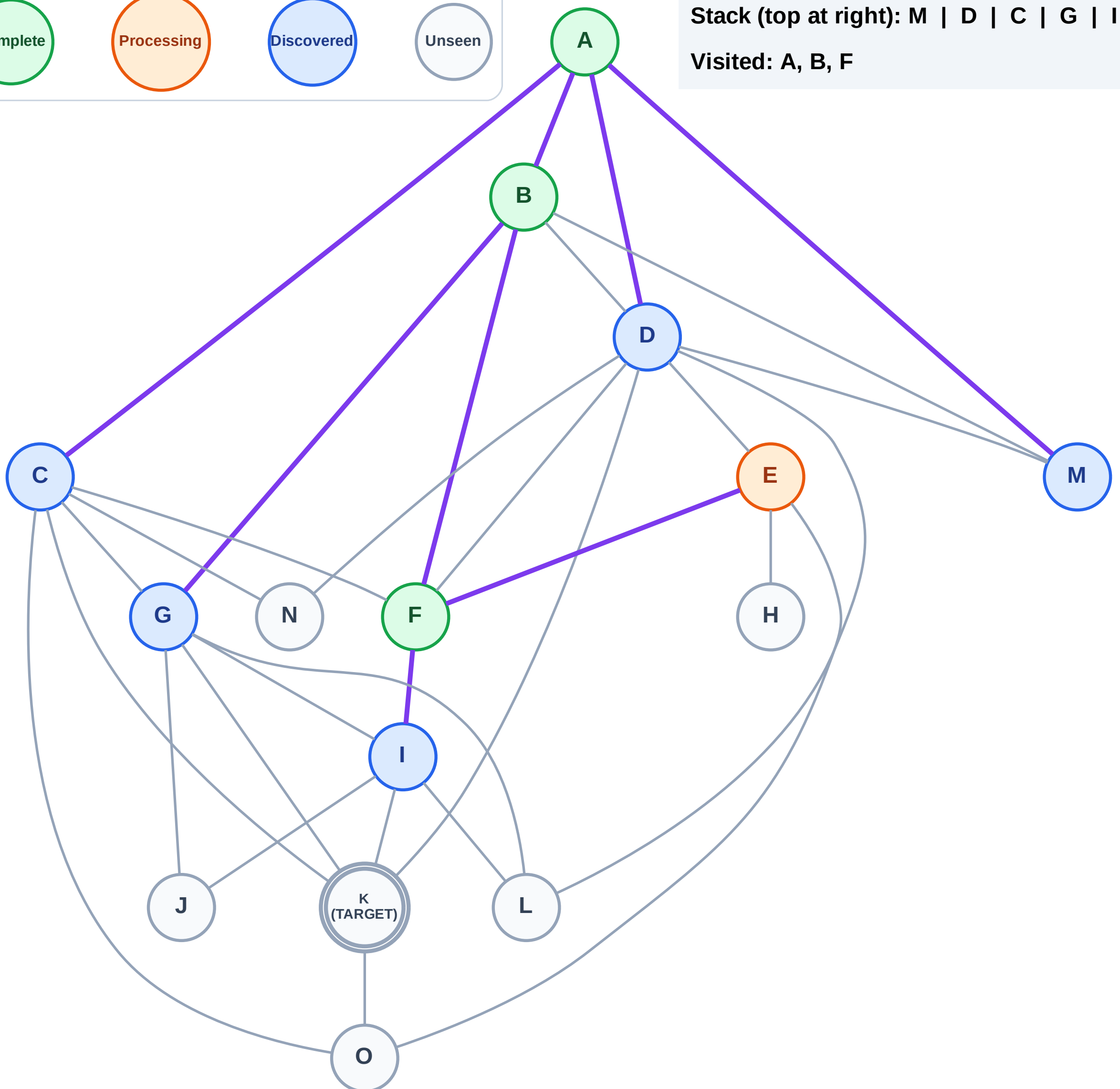
Step 8: Process node E

Legend



Stack (top at right): M | D | C | G | I

Visited: A, B, F



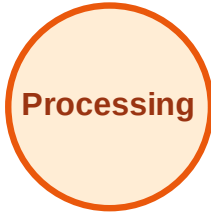
DFS Traversal

Step 9: Discover O, H from E

Legend



Complete



Processing



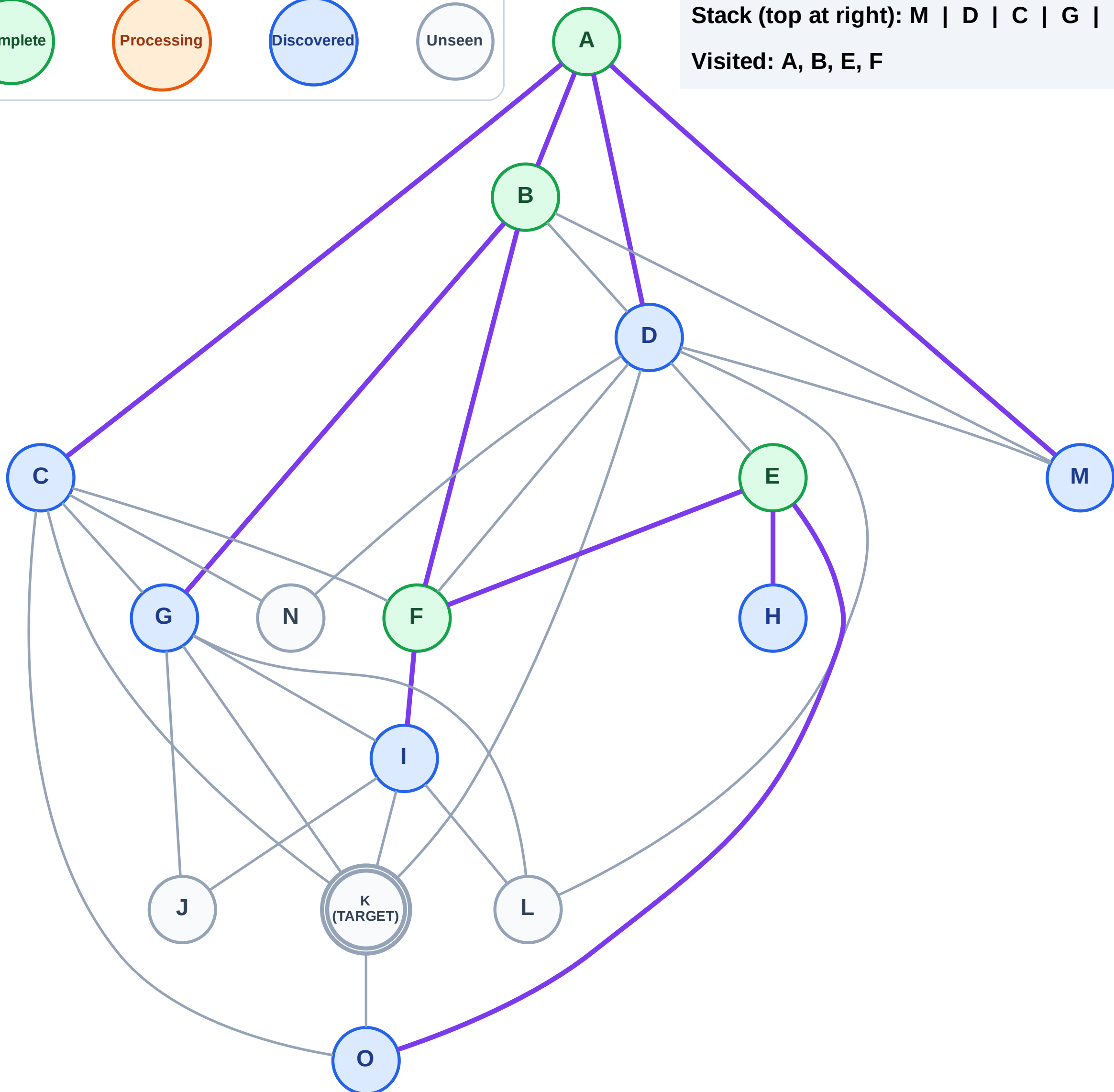
Discovered



Unseen

Stack (top at right): M | D | C | G | I | O | H

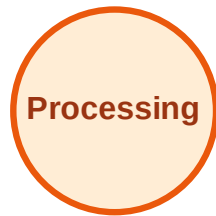
Visited: A, B, E, F



DFS Traversal

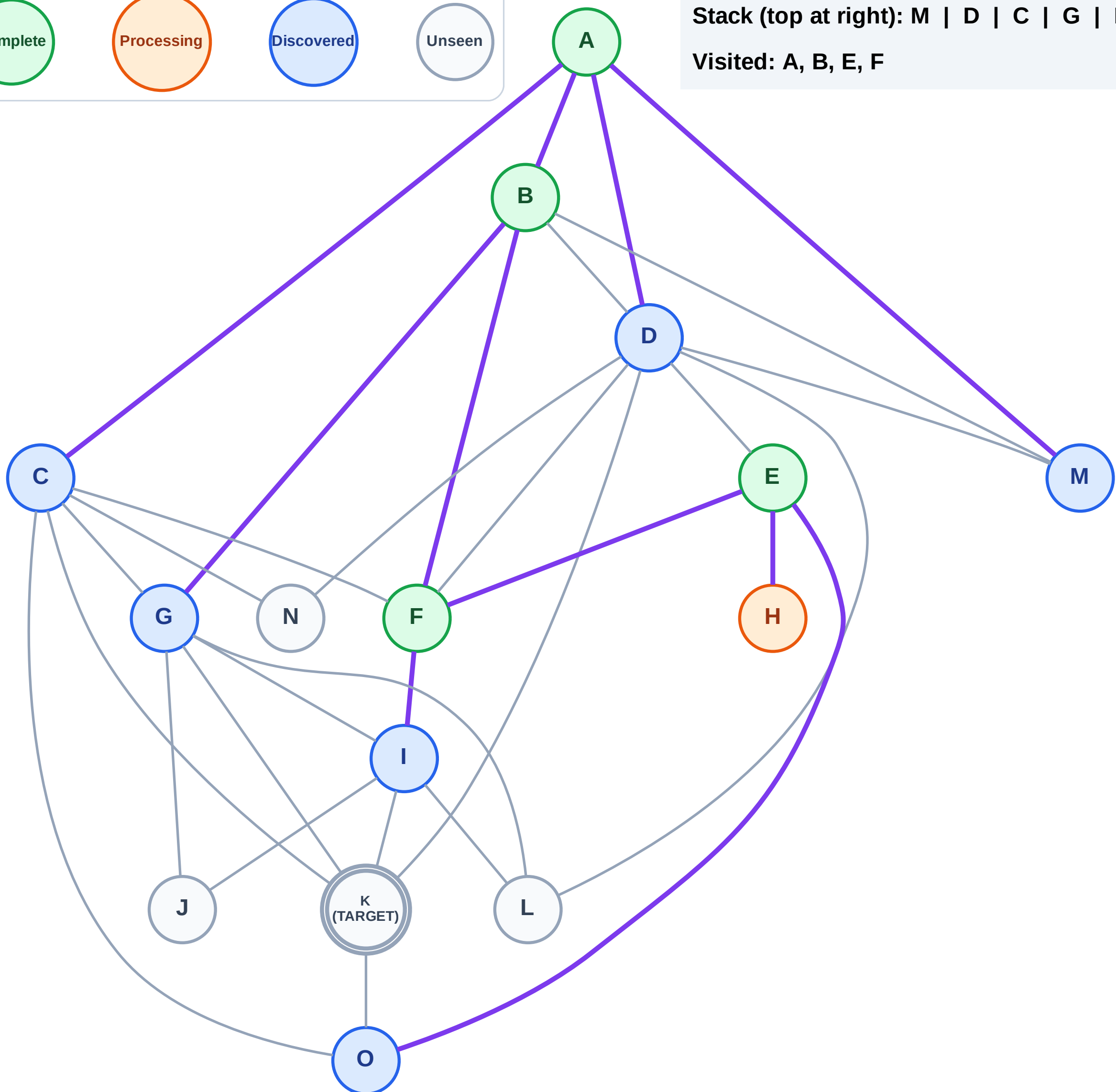
Step 10: Process node H

Legend



Stack (top at right): M | D | C | G | I | O

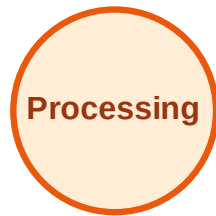
Visited: A, B, E, F



DFS Traversal

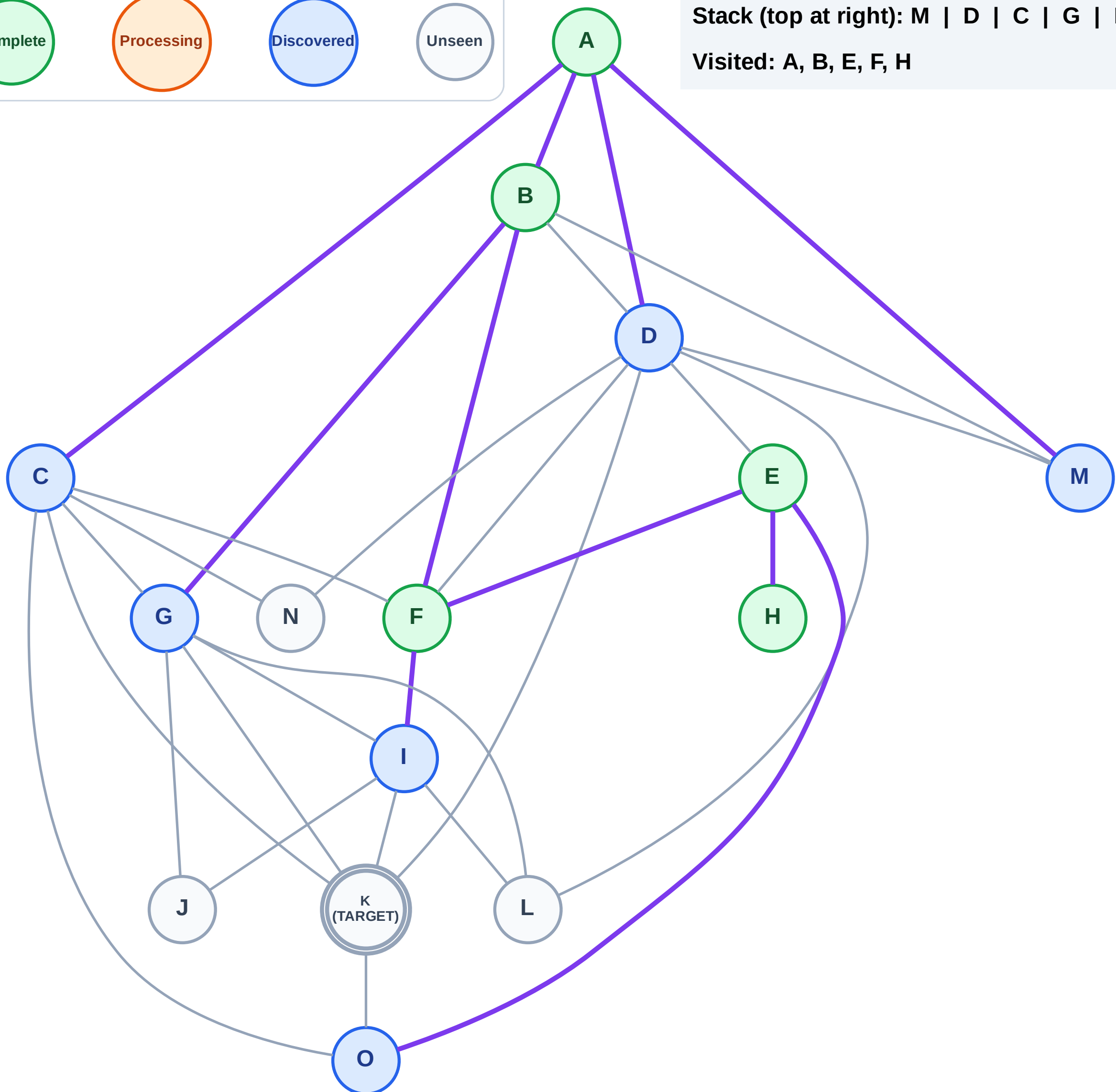
Step 11: No new neighbors from H

Legend



Stack (top at right): M | D | C | G | I | O

Visited: A, B, E, F, H



DFS Traversal

Step 12: Process node O

Legend

Complete

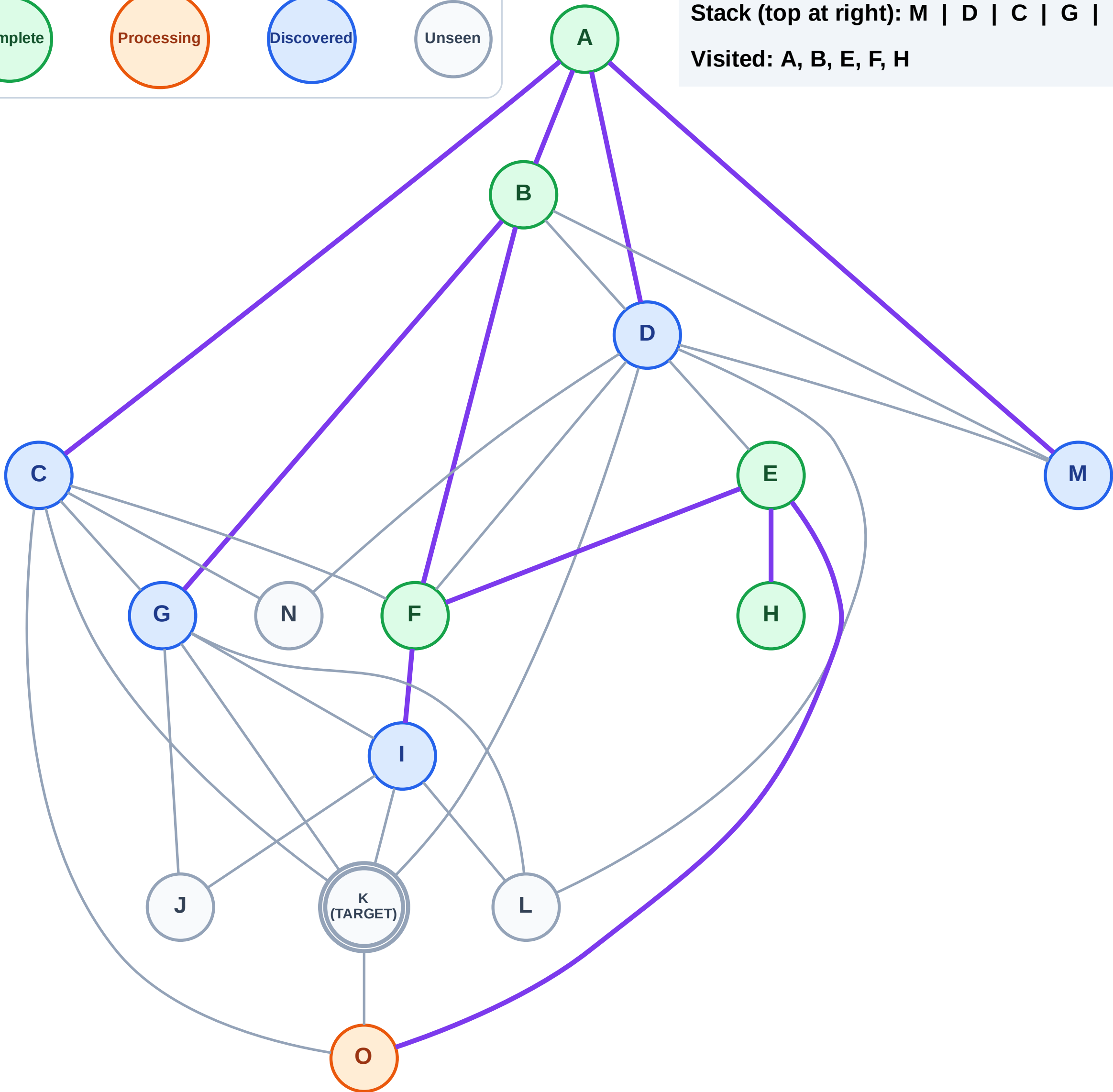
Processing

Discovered

Unseen

Stack (top at right): M | D | C | G | I

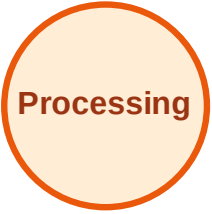
Visited: A, B, E, F, H



DFS Traversal

Step 13: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → F → E → O → K

Legend



Stack (top at right): M | D | C | G | I

Visited: A, B, E, F, H, K, O

